

i Info - dat540.22H

Exam

Faculty of Science and Technology

Course code: DAT540

Course title: Introduction to Data Science

Year and semester: 2022 Fall (Resit)

Date: 28.02.2023

Time: 3 hours

Legal aids: None

All exams submitted after the expired upload time will not be registered.

Questions - All Questions will be graded after the exam. Therefore, you will NOT receive any points immediately after completion.

- **Multiple Choice (40 questions):** You get 1 point for each correct answer to the multiple choice questions. You will get 0 points for incorrect answers or unattempted questions.
 - *NOTE: If options for some questions seem to have multiple fits, only selecting the one that fits best to the question will result in a correct evaluation.*
- **File Uploads / Essay Type Questions: NONE**

Supporting material: No Aid. You are not allowed to get help from other people when working on your exam assignment. We are also reminding you that you when registering for the semester, you signed that you have read and understood the rules for cheating and plagiarism in the Exam Rules and Regulations at the University of Stavanger. Plagiarism control will be carried out.

Important contact information during the exam: If something should happen during the exam and you need help, you can call one of the phone numbers below if you need technical support, administrative support, or have questions about the course responsible.

Course responsible:

- Antorweep Chakravorty, antorweep.chakravorty@uis.no, Phone: +47 40625773
- Neel Kanwal, neel.kanwal@uis.no, Phone: +47 51831420

Administrative support: 51 83 31 26

NB! If something goes wrong in Inspira and you cannot hand in your assignment, you need to contact administrative support ASAP.

Important Information about Exams in Inspira

If you have been given extra time on the exam, it has been added. You can see the count down at the top of the page. Here you will also find your candidate number.

Handing in

The exam will automatically close when the time is up. We recommend that you get yourself informed of how to best carry these steps out in due time and before you start working on the assignments.

1 dat540.22h.resit

Assume there is a 'Daily Demand Forecasting Orders' data set named 'daily_demand'. Now, add a column named 'Order10'. 'Order10' which will take the maximum value for 'Target(total orders)' for 10 rows at a time (previous 9 rows + current row). Which of the following implementation would retrieve 20 rows excluding the first 10 rows?

Select one alternative:

☐ `daily_demand['Order10'] = daily_demand.groupby('Target (Total orders)').max()`

☐ `daily_demand[10:30]`

☐ `daily_demand['Order10'] = daily_demand['Target (Total orders)'].sort_values().max()`

☐ `daily_demand.head(20)`

☐ `daily_demand['Order10'] = daily_demand['Target (Total orders)'].rank('max')`

☐ `daily_demand.tail(30).head(20)`

☐ `daily_demand['Order10'] = daily_demand['Target (Total orders)'].rolling(10).max()`

☐ `daily_demand.head(30).tail(20)`

Maximum marks: 1

2 dat540.22h.resit

You are given the following code where you only want to print the odd numbers. What in the blank area (BLANKAREA) would satisfy this?

```
for i in range(5):  
    if i % 2 == 0:  
        BLANKAREA  
    else:  
        print(i)
```

Select one alternative:

- ☐ continue
- ☐ pass
- ☐ Both pass and continue
- ☐ Both pass and continue including break

Maximum marks: 1

3 dat540.22h.resit

Consider a csv file (csv_data.csv) is loaded and opened. What do you think the next() would do?

```
with open(any_csv_file, 'r') as csv_data:  
    next(csv_data)  
    d = list(csv.reader(csv_data))
```

Select one alternative:

- ☐ the d variable is a list with all the file contents in csv_data.csv.
- ☐ the d variable is a list with all file contents, excluding the header.
- ☐ the d variable is a list and is writable and its contents can be changed.
- ☐ None of the above.

Maximum marks: 1

4 dat540.22h.resit

Given a numpy array

```
a = np.array([5, 2, 2, 3])
```

What would be the output of **a** after the following operation,
`a[1:] += a[:3]`

Select one alternative:

- ☐ [5, 4, 4, 6]
- ☐ [5, 7, 4, 6]
- ☐ [5, 7, 4, 5]
- ☐ [5, 6, 5, 7]

Maximum marks: 1

5 dat540.22h.resit

What do you think of the primary difference in classification and regression-based problems?

Select one alternative:

- ☐ Classification for discrete and regression for continuous value problems.
- ☐ Both are the same.
- ☐ Classification for continuous and regression for discrete value problems.
- ☐ None of these.

Maximum marks: 1

6 edat540.22h.01

For a dictionary `new_dict = {0: "airplane", 1: "bird", 2: "cat", 3:"dog"}`, Which of the following statements will display ONLY values.

1. `[ i[1] for i in new_dict.items() ]`
2. `new_dict.values()`
3. `new_dict.items()`

Select one alternative:

- ☐ 2 & 3
- ☐ 1 & 2
- ☐ All of the options
- ☐ 1 & 3

Maximum marks: 1

7 edat540.22h.02

What do you think the output of the following would be,

```
salesByRegion = pd.DataFrame({'area': ['Gosen',  
'Madlamark', 'UiS Vest', 'Centrum', 'Mariero',  
'Sandnes'],  
    'sales': [150, 250, 350, 450, 855, 858]})  
  
list(salesByRegion[::2]['area'])
```

Select one alternative:

- ☐ ['Sandnes', 'Mariero', 'Centrum']
- ☐ ['Mariero', 'Gosen', 'UiS Vest']
- ☐ ['Gosen', 'UiS Vest', 'Mariero']
- ☐ ['Gosen', 'Madlamark']

Maximum marks: 1

8 edat540.22h.02

Given the dataframe,

```
df = pd.DataFrame({'class': ['Spam', 'Ham', 'Ham']})
```

How would you categorize the Spam with 0 and Ham with 1.

Select one alternative:

- ☐ df.replace(to_replace=["Spam", "Ham"], value=[0, 1])
- ☐ df.replace(["Spam", "Ham"], [0, 1])
- ☐ All of the options
- ☐ df.loc[df['class'].isin(['Spam'])] = 0
df.loc[df['class'].isin(['Ham'])] = 1

Maximum marks: 1

9 edat540.22h.03

If the columns in a pandas dataframe are heterogenous, what will be the dtype of the values of that dataframe?

Select one alternative:

- ☐ object
- ☐ int
- ☐ None
- ☐ float

Maximum marks: 1

10 edat540.22h.04

. Observe the following code and identify what will be the outcome?

```
import pandas as pd  
s = pd.Series([1,2,3,4,5], index=list('abcde'))  
print(s['a'])
```

Select one alternative:

- ☐ 1
- ☐ 4
- ☐ None of the options
- ☐ 3

Maximum marks: 1

11 edat540.22h.02

What is the meaning of performing train-test split?

Select one alternative:

- ☐ To test the model.
- ☐ To train the model.
- ☐ To differentiate the datasets for training
- ☐ To separate data for model training and testing.

Maximum marks: 1

12 edat540.22h.01

What would be the shape and size of the initialized array using the following code?

```
arr = np.array([[[1,2],[3,4],[5,6]]], dtype=np.uint8)
```

Select one alternative:

- ☐ (3,2,1), float8
- ☐ (3,2,1), float64
- ☐ (1,3,2), uint8
- ☐ (1,3,2), float8

Maximum marks: 1

13 edat540.22h.03

Which implementation would let you load 5 rows of data from any large .csv file using 'pandas'?

Select one alternative:

- ☐ `pd.read_csv('filename.csv', chunksize=5, delimiter=',')`
- ☐ `file = pd.read_csv('filename.csv')`
`file.head()`
- ☐ `file = pd.read_csv('filename.csv')`
`file.tail()`
- ☐ `pd.read_csv('filename.csv', nrows=5)`

Maximum marks: 1

14 edat540.22h.04

Which of these options will not select the last three rows of this DataFrame?

		a	b	c	d
idx1	idx2				
bar	A	4	7	10	13
	B	5	8	11	14
foo	A	6	9	12	15
	B	16	17	18	19

Select one alternative:

- ☐ `df[('B', 'bar'):('foo', 'B')]`
- ☐ `df.loc[('foo', 'B'):'bar']`
- ☐ None of the options
- ☐ `df.iloc[(1, 1): (0, 1)]`

Maximum marks: 1

15 edat540.22h.03

We have a Pandas Series named 'pd_sr' of 100 random normally distributed numbers. And a 'lambda function' named 'f' as follows:

```
f = lambda x: round(x % 3, 2)
```

Which of the following is not a valid method of implementing this function on our defined Series, pd_sr ?

Select one alternative:

- ☐ pd_sr.apply(f)
- ☐ All of the options
- ☐ pd_sr.applymap(f)
- ☐ pd_sr.map(f)

Maximum marks: 1

16 edat540.21h

In order to convert a timezone to another timezone, which function should be used? What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ tz_convert()
- ☐ tz_localize()
- ☐ convert()
- ☐ localize()

Maximum marks: 1

17 edat540.21h

Moving window operations in pandas can be performed by using which function? What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ rolling()
- ☐ mean()
- ☐ movingWindowFunction()
- ☐ All of the options

Maximum marks: 1

18 edat540.21h

When we call the **describe()** method for a **grouped** DataFrame of numeric column values, what are the reported parameters? What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ Count
- ☐ Mean, STD
- ☐ Min, Max
- ☐ All of the options

Maximum marks: 1

19 edat540.21h

Given the following code snippet

```
import pandas as pd
import numpy as np
np.random.seed(1234)
# Generate two data sets with random values.
df1 = pd.DataFrame({'key': ['b', 'b', 'a', 'c', 'a', 'a', 'd'],
                    'data1': np.random.randint(0, 10, 7)})
df2 = pd.DataFrame({'key': ['a', 'b', 'd'],
                    'data2': np.random.randint(0, 20, 3)})
print('df1:\n', df1)
print('df2:\n', df2)
```

If we execute **pd.merge(df1, df2)**, what is the result that we get?

What will be the **best** option that covers all possibilities

A.

	key	data1	data2
0	b	3	11
1	b	6	11
2	a	5	9
3	a	8	9
4	a	9	9
5	d	1	12

B.

	key	data1	data2
0	b	3	11.0
1	b	6	11.0
2	a	5	9.0
3	a	8	9.0
4	a	9	9.0
5	c	4	NaN
6	d	1	12.0

C.

	key	data1	data2
0	b	3	11.0
1	b	6	11.0
2	a	5	9.0
3	c	4	NaN
4	a	8	9.0
5	a	9	9.0
6	d	1	12.0

Select one alternative:

- ☐ A
- ☐ B
- ☐ C
- ☐ None of the options

Maximum marks: 1

20 edat540.21h

Given the following code snippet

```
import numpy as np
arr = np.array([[1,2,3], [4,5,6], [7,8,9]])
```

What will we retrieve if we use **arr[[False, True, False]]**

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ None
- ☐ array([])
- ☐ array([[4, 5, 6]])
- ☐ None of the options

Maximum marks: 1

21 edat540.21h

Palindrome is a string that has the same sequence of characters forward and in reverse. For example 'HannaH'. This string will have the same sequence of characters if we write it in reverse as well. However for another string 'abc', if we write it in reverse, it will be 'cba' and therefore won't be a palindrome

Lets assume we have string

s = 'aba'

We want to check (True) if it is a palindrome. What will be the best way to check just this case where s='aba'?

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ s == s[::-1]
- ☐ s == s[-1::-1]
- ☐ s == s[2::-1]
- ☐ All of the options

Maximum marks: 1

22 edat540.21h

We want to **cast** a variable to type **int**. What will happen if we pass a value such as '**12.e2**'

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ 12
- ☐ ValueError
- ☐ '12.e2'
- ☐ None of the option

Maximum marks: 1

23 edat540.21h

Given the following code snippet

```
import pandas as pd
import numpy as np
from datetime import datetime as dt
dates = pd.date_range('2021-10-01 10:00', periods=100, freq='T')
s = pd.Series(np.random.randn(100), index=dates)
```

We would like to shift the values forward by 2. What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ None of the options
- ☐ s.shift(2, freq='T')
- ☐ s.shift(-2)
- ☐ s.shift(2)

Maximum marks: 1

24 edat540.21h

What does the **add_patch()** for a plot do?

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ Adds a subplot
- ☐ Can be used to plot shapes like a rectangle, circle
- ☐ Plots a sequence of values as histogram
- ☐ Plots a sequence of values as line plot

Maximum marks: 1

25 edat540.21h

The pandas function **get_dummies()** is used for a process called? What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ Scaling
- ☐ Standardization
- ☐ One Hot Encoding
- ☐ None of the options

Maximum marks: 1

26 edat540.21h

pandas is a module in python that allows tabular representation of data

Select one alternative:

- ☐ False
- ☐ True

Maximum marks: 1

27 edat540.21h

Given the following code snippet:

```
import numpy as np
arr = np.random.randint(0, 100, 40).reshape(5, 2, 4)
```

What would be the shape of **arr.transpose(2, 0, 1)**

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ (4, 2, 5)
- ☐ (2, 0, 1)
- ☐ (5, 2, 4)
- ☐ (4, 5, 2)

Maximum marks: 1

28 edat540.21h

Given the code snippet

```
x = 1
lst = []
while True:
    if x % 5 == 0:
        break
    lst.append(x)
    x += 1
print(lst)
```

What will be printed for **lst**? What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ [0, 1, 2, 3, 4]
- ☐ [1, 2, 3, 4, 5]
- ☐ []
- ☐ [1, 2, 3, 4]

Maximum marks: 1

29 edat540.21h

Out of the following data structures in python, which is **immutable**?

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ Dict
- ☐ Set
- ☐ List
- ☐ Tuple

Maximum marks: 1

30 edat540.22h.04

Which code produces this output: {'a', 'b', 'c'}, in no particular order

Select one alternative:

- ☐ None of the options
- ☐ List("abc")
- ☐ Tuple("abc")
- ☐ set("cbab")

Maximum marks: 1

31 edat540.22h.04

Observe the following code and identify what will be the outcome?

```
tools = ["Pandas", "Matplotlib", "Numpy"]

for x, y in enumerate(tools):
    print("{}: {}".format(x, y), end = " ")
```

Select one alternative:

- ☐ Pandas, :Matplotlib, :Numpy,
- ☐ Exception
- ☐ 0: Pandas, 1: Matplotlib, 2: Numpy,
- ☐ 0: , 1: , 2: ,

Maximum marks: 1

32 edat540.22h.04

Observe the following code and identify what will be the outcome?

```
a = ['DAT', 'STA', 'ELE']  
'3'.join(a)
```

Select one alternative:

- ☐ '3DATSTAELE'
- ☐ 'DAT3STA3ELE3'
- ☐ '3DAT3STA3ELE'
- ☐ 'DAT3STA3ELE'

Maximum marks: 1

33 edat540.21h

Given the following code snippet

```
import numpy as np  
arr = np.array([1,2,3], [4,5,6], [7,8,9])
```

What value would **arr[0,1]** retrieve?

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ 2
- ☐ 1
- ☐ [1,2,3]
- ☐ None of the options

Maximum marks: 1

34 edat540.21h

Given the following code snippet

```
import numpy as np
# Creating a multi dimensional array for a nested python list
arr = np.random.random(100)
arr.reshape(10,10)
```

What is the shape of **arr** after executing the above snippet?

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ (10, 10)
- ☐ (10,)
- ☐ (,10)
- ☐ (100,)

Maximum marks: 1

35 edat540.21h

Given the following code snippet

```
import pandas as pd
import numpy as np
s = pd.Series(np.arange(100))
```

We want to print the **last 5** records in this series.

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ All of the options
- ☐ `s.tail(5)`
- ☐ `s[len(s)-5:]`
- ☐ `s[-5:]`

Maximum marks: 1

36 edat540.21h

Given a DataFrame **df**, we would like to add a new column named **cn**. We assume that there are in total 5 rows in the dataframe. What would be the best way to do this?

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ `df['cn'] = np.random.random(5)`
- ☐ `df('cn') = np.random.random(5)`
- ☐ `df[cn] = np.random.random(5)`
- ☐ `df.cn = np.random.random(5)`

Maximum marks: 1

37 edat540.21h

There is a numpy array that we want to store to a file. We can write it using the numpy function

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ read_csv
- ☐ load
- ☐ open
- ☐ None of the options

Maximum marks: 1

38 edat540.21h

What will be the output of the following

list(range(5, -1))

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ [4, 3, 2, 1, 0]
- ☐ [-1, 0, 1, 2, 3, 4]
- ☐ []
- ☐ [5, 4, 3, 2, 1, 0]

Maximum marks: 1

39 edat540.21h

Given the following code snippet

```
lst1 = [1, 2, 3]  
lst2 = [4, 5]
```

We want to join both of the list. We use a **function** and get the result as: **[1, 2, 3, 4, 5]**

Which function have we used?

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ append
- ☐ join
- ☐ merge
- ☐ extend

Maximum marks: 1

40 edat540.21h

Given the following code snippet

```
import numpy as np
arr = np.array([-1, 2, 4, 0, 2, -1, 1, 0])
np.sort(arr)
```

In what order are the values in **arr** stored after executing the snippet

What will be the **best** option that covers all possibilities

Select one alternative:

- ☐ array([-1, -1, 0, 0, 1, 2, 2, 4])
- ☐ array([-1, 2, 4, 0, 2, -1, 1, 0])
- ☐ array([4, 2, 2, 1, 0, 0, -1, -1])
- ☐ None of the options

Maximum marks: 1